Problem 3

$A_1 = [[1,0],[0,0]]$

$A_2 = [[0,1],[0,0]]$

$A_3 = [[0,0],[1,0]]$

$A_4 = [[0,0],[0,1]]$

Part 1 Take an arbitarary 2x2 Matrix

$M = [[a,b],[c,d]]$ where $a,b,c,d \in R$.

We need scalars K_1, K_2, K_3, K_4 such that

$K_1A_1 + K_2A_2 + K_3A_3 + K_4A_4 = M$.

Computing the left side:

$K_1A_1 + K_2A_2 + K_3A_3 + K_4A_4 = [[K_1,K_2],[K_3,K_4]]$

Setting this equal to M and comparing entries

$K_1 = a$

$K_2 = b$

$K_3 = c$

$K_4 = d$

This system always has a solution, whatever a,b,c,d are. So every matrix in M can be written as

$M = aA_1 + bA_2 + cA_3 + dA_4$

Hence $\text{span}(S) = M_{22}$.

Express B as a linear combination of S

$B = [[3, -2], [5, 4]]$

Set up

$K_1A_1 + K_2A_2 + K_3A_3 + K_4A_4 = [[K_1, K_2], [K_3, K_4]]$

$= [[3, -2], [5, 4]]$.

Comparing entries

$K_1 = 3$

$K_2 = -2$

$K_3 = 5$

$K_4 = 4$

Result: $B = 3A_1 - 2A_2 + 5A_3 + 4A_4$

Problem 4

Part 1 $x: C_1 + 3C_2 + 2C_3 = 0$

$y: 2C_1 + C_2 - C_3 + 5C_4 = 0$

$z: -C_1 + 4C_2 + 5C_3 + 2C_4 = 0$

add Eq 1 and Eq 3:

$7C_2 + 7C_3 + 2C_4 = 0$

Eq 2 -2Eq 1 : $-5C_2 - 5C_3 + 5C_4 = 0 \Rightarrow C_4 = C_2 + C_3$

Subsituting : $7(C_2 + C_3) + 2(C_2 + C_3) = 9(C_2 + C_3) = 0$

so $C_2 = -C_3$ and $C_4 = 0$

Eq 1 gives : $C_1 = 3C_2 - 2C_3 = 3C_3 - 2C_3 = C_3$.

Taking $c_3 = t$: $(c_1, c_2, c_3, c_4) = (t, -t, t, 0)$ which is non trivial for $t \neq 0$.

Conclusion: V is linearly dependent.

Part2: $t = 1$

$v_1 - v_2 + v_3 = 0$

$v_3 = v_2 - v_1$

$(3,1,4) - (1,2,-1) = (2,-1,5) = v_3$ ✓

$v_1 = v_2 - v_3$ and $v_2 = v_1 + v_3$.

Part3: $W = \{(1,0,0), (0,1,0), (a,b,c)\}$

Suppose

$\alpha(1,0,0) + \beta(0,1,0) + \gamma(a,b,c) = (0,0,0)$

component-wise

$\alpha + \gamma a = 0$

$\beta + \gamma b = 0$

$\gamma c = 0$

Case $c \neq 0$: Then $\gamma = 0$, $\alpha = 0$, $\beta = 0$, so W is independent

Case $c = 0$: Take $\gamma = 1$, $a = -a$, $\beta = -b$, non trivial solution. Then $(a,b,0) = ae_1 + be_2$ lies in the Span of the first two vectors, so W is dependent.

Equivalently, the determinant of the matrix with columns of W is

$$\begin{vmatrix} a & 0 \\ 1 & a \end{vmatrix}$$

$$\begin{vmatrix} 0 & b \\ 1 & b \end{vmatrix} = c$$

$$\begin{vmatrix} 0 & c \\ 0 & c \end{vmatrix}$$

which is non zero exactly when $c \neq 0$.
W is linearly independent if $c \neq 0$, with a and b any real numbers.

Problem 5

Part 1 ~~Write~~ $A = C_1T_1 + C_2T_2 + C_3T_3 + C_4T_4$

$= [[C_1, C_2], [C_3, C_4]]$.

$C_1 = 3,\ C_2 = 1,\ C_3 = 2,\ C_4 = 4$.

$[A] - B = (3, 1, 2, 4)^T$

Part 2: $aS_1 + bS_2 = [[a+b, a-b], [0,0]] = [[1,0],[0,0]]$.

Then $a+b = 1$ and $a-b = 0$, so $a = b = 1/2$.

$T_1 = 1/2\ S_1 + 1/2\ S_2$, so $[T_1]_B' = (1/2, 1/2, 0, 0)$.

$T_2 = 1/2\ S_1 - 1/2\ S_2$, $[T_2]_B' = (1/2 - 1/2, 0, 0)$

$T_3 = 1/2\ S_3 + 1/2 S_4$, So $[T_2]_B' = (0, 0, 1/2, 1/2)$.

$T_4 = 1/2\ S_3 - 1/2\ S_4$, so $[T_4]_B' = (0, 0, 1/2, -1/2)$.

Transition matrix:

$$P_{B \to B'} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

Part B: Coordinate vector $[A]_{B'}$

Use $[A]_{B'} = P\{B \to B'\}[A]_B$:

$$[A]_{B'} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \\ 2 \\ 4 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 4 \\ 2 \\ 6 \\ -2 \end{bmatrix}$$

$$[A]_{B'} = (2, 1, 3, -1)^T$$

Problem 6

Part a) $R_1: x_1 + 2x_2 - x_3 + 3x_4 - 2x_3 = 0$

$R_2: 2x_1 + 4x_2 - 2x_3 + 7x_4 - 3x_5 = 0$

$R_3: 3x_1 + 6x_2 - 3x_3 + 10x_4 - 5x_5 = 0$

Row Reduce:

$R_2 - 2R_1: x_4 + x_5 = 0$

$R_3 - 3R_1: x_4 + x_5 = 0$

$R_1 - 3(x_4 + x_5 = 0): x_1 + 2x_2 - x_3 - 5x_5 = 0$

RREF System:

$$x_1 + 2x_2 - x_3 - 5x_5 = 0$$

$$x_4 + x_5 = 0$$

$$x = s\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + r\begin{bmatrix} 5 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Basis: $\{(-2,1,0,0,0), (1,0,1,0,0), (5,0,0,-1,1)\}$

Part a.2)

rank = 2, dim = 5-3 = 3

Part a.3)

$$x_1 + x_2 + x_3 = 0$$

$$x_1 = -2x_2 + x_3 + 5x_5$$

$$x_2 = 2x_3 + 5x_5$$

$$x_3 = t, \quad x_5 = r$$

$$x_2 = 2t + 5r$$

$$x_1 = -2(2t + 5r) + t + 5r = -3t - 5r$$

$$x_4 = -r$$

$$x = t\begin{bmatrix} -3 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + r\begin{bmatrix} -5 \\ 5 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

New basis: $\{(-3,2,1,0,0), (-5,5,0,-1,1)\}$, new dimension = 2

Partb

b.1 : Symmetric 3x3 matrices

$W = \{A \in M_{33} : A = A^T\}$

Subspace proof :

1) Zero: $0^T = 0$, so $0 \in W$
2) if $A, B \in W$, then $(A+B)^T = A^T + B^T = A+B$, so $A+B \in W$.
3) if $A \in W$ and $k \in R$, then $(kA)^T = kA^T = kA$, so $kA \in W$.

Hence W is a subspace of M_{33}.

Part b.2 Polynomials in P_4 with $P(1) = P(-1) = 0$

Let $U = \{P \in P_4 : P(1) = 0$ and $P(-1) = 0\}$, where P_4 is the space of polynomials of degree at most 4 (dim 5).

Subspace proof.

1) Zero: The zero polynomial satisfies $0(1) = 0(-1) = 0$
2) Addition : if $P, Q \in V$, then $(P+Q)(1) = 0+0 = 0$ and $(-1)(P+Q) = 0+0 = 0$.
3) $(kP)(1) = k \cdot (0) = 0$ and $kP(-1) = 0$.

Hence U is a subspace of P_4.

Dimension and basis : By the factor theorem, $P(1) = P(-1) = 0$ means $(x-1)(x+1) = x^2 - 1$ divides p. So.

$P(x) = (x^2-1)\cdot q(x)$ where q has degree ≤ 2

So $q = a + bx + cx^2$.

$P(x) = a(x^2-1) + b(x^3-x) + c(x^4-x^2)$.

Basis : $\{x^2-1, x^3-x, x^4-x^2\}$

These are independent since they have distinct lower or highest degree terms (degree 2, 3, 4) and they Span U by the factorisation above.

dim $U = 3$, .

Problem 7

Part 1)

$\varphi = [u_1', u_2'] = [[1,-1], [3,-1]]$ Then

$\det \varphi = -1+3 = 2$ and

$\varphi^{-1} = 1/2\ [[-1,1], [-3,1]]$

$[u_1]_{B'} = \varphi^{-1}(2,2)^T = 1/2\,(-2+2, -6+2) = (0,-2)^T$

$[u_2]_{B'} = \varphi^{-1}(4,-1)^T = 1/2\,(-4, -11, -12, -1) = (-5/2, -13/2)^T$

$$P_{B \to B'} = \begin{bmatrix} 0 & -5/2 \\ -2 & -13/2 \end{bmatrix}$$

Part2) use $[w]_{B'} = P\{B \to B'\}[w]_B$:

$[w]_{B'} = (-5, -11)^T$

$$[w]_{B'} = \begin{bmatrix} 0 & -5/2 \\ -2 & -13/2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -5 \\ 2 -13 \end{bmatrix}$$

Problem 8

~~$R_2 = 2P$~~

- $R_2 - 2R_1 = (0,0,1,3,-2,-6)$
- $R_3 - 2R_1 = (0,0,1,3,-1,-1)$
- $R_4 + R_1 = (0,0,0,0,0,0)$
- $(R_3 - 2R_1) - (R_2 - 2R_1) = (0,0,0,0,1,5)$

$$R = \begin{bmatrix} 1 & -3 & 0 & -14 & 0 & -37 \\ 0 & 0 & 1 & 3 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot cols = 1,3,5. Free var = x_2, x_4, x_6, Rank=3

Part 1)

$\{(1,-3,0,-14,0,-37), (0,0,1,3,0,4), (0,0,0,1,5)$

$\dim^{row}(A) = 3.$

Part2) $\{(1,2,2,-1)^T, (4,9,9,-4)^T, (5,8,9,-5)^T\}$

$\dim \text{col}(A) = 3$.

Part3)

From RREF : $x_2 = s$, $x_4 = t$, $x_6 = u$:

1-) $x_1 = 3s + 14t + 37u$

2-) $x_3 = -3t - 4u$

3-) $x_5 = -5u$

$$x = s\begin{bmatrix}3\\1\\0\\0\\0\\0\end{bmatrix} + t\begin{bmatrix}14\\0\\-3\\1\\0\\0\end{bmatrix} + u\begin{bmatrix}37\\0\\-4\\0\\-5\\1\end{bmatrix}$$

Basis : $\{(3,1,0,0,0,0), (14,0,-3,1,0,0),$
$(37,0,-4,0,-5,1)\}$, so nullity $= 3$.

Part4) No. every column of A has its 4th entry equal to the negative of its 1st entry (since $R_4 = -R_1$).

So every vector in $\text{col}(A)$ satisfies $y_4 = -y_1$.

~~For b : $y_1 = 1$ and~~ For b: $y_{4_1} = 1$ and $y_4 = 4$

but $-y_1 = -1 \neq 4$.

Problem 9

Part 1)

rank (A) $\leq$ min (5,7) = 5.

Largest possible rank = 5

By Rank - Nullity Theorem, rank(A) + nullity(A) = 7

so nullity (A) = 7 - rank (A) $\geq$ 7 - 5 = 2

Smallest possible Nullity = 2.

Part 2)

freev Variables = 7 - rank(A) = 7 - 4 = 3

Because the system is consistent, the general solution is $X_P + s \cdot n_1 + t \cdot n_2 + u \cdot n_3$, where X_P is a particular solution and $\{n_1, n_2, n_3\}$ is a basis for null(A). So there are three parameters.

b) No. null space dimension = 3, so it contains nonzero vectors. For any non zero n in null(A), $X_P + n$ is also a solution, since $A(X_P + n) = b + 0 = b$.

Hence the system has infinitely many solutions.

Part 3

(a) Rank of A

By Rank-Nullity : rank(A) = 7 - nullity(A) =

7 - 2 = 5

(b) Yes the column space col(A) is a subspace of R^5 with dimension 5 is R^5 itself, so $col(A) = R^5$

Equivalently, A has a pivot in every one of its 5 rows so $Ax = b$ is consistent for every $b \in R^5$. The 7 column span R^5.